# P2 Chapter 11: Integration

## Solving Differential Equations

# Introduction

1 Solve $\frac{dy}{dx} = x$

2 Solve $\frac{dy}{dx} = -\frac{x}{y}$

3 Solve $\frac{dy}{dx} = y$

4 Solve $\frac{dE}{dv} = mv$

4 Solve $\frac{dE}{dx} = kx$

# Differential Equations

**Differential equations are equations involving a mix of variables and derivatives, e.g. $y$, $x$ and $\frac{dy}{dx}$.**

'<u>Solving</u>' these equations means to get $y$ in terms of $x$ (with no $\frac{dy}{dx}$).

**Q** Find the general solution to $\frac{dy}{dx} = xy + y$

?

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**STEP 1:** Get $y$ to the side of $\frac{dy}{dx}$ by dividing and $x$ to the other side.
(you may need to factorise to separate out $y$ first)

**STEP 2:** Integrate both sides with respect to $x$. $\frac{dy}{dx}\,dx$ simplifies to $dy$ (recall that (implicitly) differentiating an expression in terms of $y$ with respect to $x$ introduces a $\frac{dy}{dx}$, so integrating similarly would get rid of it)

**STEP 3:** Make $y$ the subject, if the question asks.

**Fro Tip**: When you have $e^{f(x)+c} = e^{f(x)}e^{c}$, it's typical to write $Ae^{f(x)}$ where $A = e^{c}$ (noting that $A$ is any positive constant since $e^{c} > 0$)

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**Q** Find the general solution to $\frac{dy}{dx} = xy + y$

$$\frac{dy}{dx} = y(x+1)$$

$$\frac{1}{y}\frac{dy}{dx} = x + 1$$

I usually skip this line. →

$$\int \frac{1}{y}\frac{dy}{dx}\,dx = \int x + 1\ dx$$

$$\int \frac{1}{y}\ dy = \int x + 1\ \ dx$$

$$\ln|y| = \frac{1}{2}x^2 + x + C$$

$$y = e^{\frac{1}{2}x^2+x+C} \quad = e^{\frac{1}{2}x^2+x}e^C$$

$$= Ae^{\frac{1}{2}x^2+x}$$

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**STEP 2b:** If possible, try to combine your constant of integration with other terms
(e.g. by letting $C = \ln k$ where $k$ is another constant)

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$$\frac{1}{\tan y}\frac{dy}{dx} = \frac{x}{1+x^2}$$

**STEP 1:** Get $y$ to the side of $\frac{dy}{dx}$ by dividing and $x$ to the other side.
(you may need to factorise to separate out $y$ first)

$$\int \cot y \; dy = \int \frac{x}{1+x^2}\,dx$$

$$\ln|\sin y| = \frac{1}{2}\ln|1+x^2| + C$$

$$\ln|\sin y| = \frac{1}{2}\ln|1+x^2| + \ln k$$

$$\ln|\sin y| = \ln\left|k\sqrt{1+x^2}\right|$$

$$\sin y = k\sqrt{1+x^2}$$

$$y = \arcsin\left(k\sqrt{1+x^2}\right)$$

**STEP 2:** Integrate both sides with respect to $x$.

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# Differential Equations with Boundary Conditions

**Q** [Textbook] Find the general solution to $\frac{dy}{dx} = -\frac{3(y-2)}{(2x+1)(x+2)}$

Given that $x = 1$ when $y = 4$. Leave your answer in the form $y = f(x)$

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# Differential Equations with Boundary Conditions

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$$\int \frac{1}{y-2}\,dy = \int \frac{-3}{(2x+1)(x+2)}\,dx$$

Use partial fractions to split up RHS.

$$\frac{-3}{(2x+1)(x+2)} \equiv \frac{A}{2x+1} + \frac{B}{x+2}$$

$$A = -2, B = 1$$

$$\int \frac{1}{y-2}\,dy = \int \frac{1}{x+2} - \frac{2}{2x+1}\,dx$$

$$\ln|y-2| = \ln|x+2| - \ln|2x+1| + \ln k$$

$$\ln|y-2| = \ln\left|\frac{k(x+2)}{2x+1}\right|$$

$$y - 2 = \frac{k(x+2)}{2x+1}$$

When $x = 1, y = 4$: $4 - 2 = \frac{k(1+2)}{2+1} \quad \rightarrow k = 2$

$$\boldsymbol{y = 3 + \frac{3}{2x+1}}$$

**Tip:** While it doesn't matter when you determine your constant using the boundary conditions, usually it's easiest to determine it as soon as possible.

# Test Your Understanding

Edexcel C4 Jan 2012 Q4

Given that $y = 2$ at $x = \frac{\pi}{4}$, solve the differential equation

$$\frac{dy}{dx} = \frac{3}{y\cos^2 x}.$$

**(5)**

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| | | | |
|---|---|---|---|
| $\int y\,dy = \int \frac{3}{\cos^2 x}\,dx$ <br> $= \int 3\sec^2 x\,dx$ | Can be implied. Ignore integral signs | B1 | |
| $\frac{1}{2}y^2 = 3\tan x \quad (+C)$ | | M1 A1 | |
| $y = 2, x = \frac{\pi}{4}$ <br> $\frac{1}{2}2^2 = 3\tan\frac{\pi}{4} + C$ | | M1 | |
| Leading to <br> $C = -1$ | | | |
| $\frac{1}{2}y^2 = 3\tan x - 1$ | or equivalent | A1 | (5) |
| | | | [5] |

# Key Points on Differential Equations

- Get $y$ on to LHS by dividing (possibly factorising first).
- If after integrating you have $ln$ on the RHS, make your constant of integration $\ln k$.
- Be sure to combine all your $ln$'s together just as you did in C2. E.g.:

$$2\ln|x+1| - \ln|x| \quad \rightarrow \quad \ln\left|\frac{(x+1)^2}{x}\right|$$

- Sub in boundary conditions to work out your constant – better to do sooner rather than later.
- Exam questions ♥ partial fractions combined with differential equations.

# Exercise 11.7

Pearson Pure Year 2
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**Extension:**

**(STEP I 2013 Q7)**

(i) Use the substitution $y = ux$, where $u$ is a function of $x$, to show that the solution of the differential equation

$$\frac{dy}{dx} = \frac{x}{y} + \frac{y}{x} \qquad (x > 0,\ y > 0)$$

that satisfies $y = 2$ when $x = 1$ is

$$y = x\sqrt{4 + 2\ln x} \qquad (x > e^{-2}).$$

(ii) Use a substitution to find the solution of the differential equation

$$\frac{dy}{dx} = \frac{x}{y} + \frac{2y}{x} \qquad (x > 0,\ y > 0)$$

that satisfies $y = 2$ when $x = 1$.

(iii) Find the solution of the differential equation

$$\frac{dy}{dx} = \frac{x^2}{y} + \frac{2y}{x} \qquad (x > 0,\ y > 0)$$

that satisfies $y = 2$ when $x = 1$.

Solutions: ?

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$$\frac{dy}{dx} = \frac{x^2}{y} + \frac{2y}{x} \qquad (x > 0,\ y > 0)$$

that satisfies $y = 2$ when $x = 1$.

Solutions: (ii) $y = x\sqrt{6x^2 - 2x}$ for $x > \frac{1}{3}$

# Homework Exercise

**1** Find general solutions to the following differential equations. Give your answers in the form $y = f(x)$.

**a** $\frac{dy}{dx} = (1 + y)(1 - 2x)$

**b** $\frac{dy}{dx} = y \tan x$

**c** $\cos^2 x \frac{dy}{dx} = y^2 \sin^2 x$

**d** $\frac{dy}{dx} = 2e^{x-y}$

**2** Find particular solutions to the following differential equations using the given boundary conditions.

**a** $\frac{dy}{dx} = \sin x \cos^2 x$; $y = 0$, $x = \frac{\pi}{3}$

**b** $\frac{dy}{dx} = \sec^2 x \sec^2 y$; $y = 0$, $x = \frac{\pi}{4}$

**c** $\frac{dy}{dx} = 2\cos^2 y \cos^2 x$; $y = \frac{\pi}{4}$, $x = 0$

**d** $\sin y \cos x \frac{dy}{dx} = \frac{\cos y}{\cos x}$, $y = 0$, $x = 0$

**3** **a** Find the general solution to the differential equation

$x^2 \frac{dy}{dx} = y + xy$, giving your answer in the form $y = g(x)$.

**b** Find the particular solution to the differential equation that satisfies the boundary condition $y = e^4$ at $x = -1$.

**Hint** Begin by factorising the right-hand side of the equation.

**4** Given that $x = 0$ when $y = 0$, find the particular solution to the differential equation

$(2y + 2yx)\frac{dy}{dx} = 1 - y^2$, giving your answer in the form $y = g(x)$. **(6 marks)**

# Homework Exercise

**5** Find the general solution to the differential equation $e^{x+y}\frac{dy}{dx} = 2x + xe^{y}$, giving your answer in the form $\ln|g(y)| = f(x)$. **(6 marks)**

**6** Find the particular solution to the differential equation $(1 - x^2)\frac{dy}{dx} = xy + y$, with boundary condition $y = 6$ at $x = 0.5$. Give your answer in the form $y = f(x)$. **(8 marks)**

**7** Find the particular solution to the differential equation $(1 + x^2)\frac{dy}{dx} = x - xy^2$, with boundary condition $y = 2$ at $x = 0$. Give your answer in the form $y = f(x)$. **(8 marks)**

**8** Find the particular solution to the differential equation $\frac{dy}{dx} = xe^{-y}$, with boundary condition $y = \ln 2$ at $x = 4$. Give your answer in the form $y = f(x)$. **(8 marks)**

**9** Find the particular solution to the differential equation $\frac{dy}{dx} = \cos^2 y + \cos 2x \cos^2 y$, with boundary condition $y = \frac{\pi}{4}$ at $x = \frac{\pi}{4}$. Give your answer in the form $\tan y = f(x)$. **(8 marks)**

**10** Given that $y = 1$ at $x = \frac{\pi}{2}$, solve the differential equation $\frac{dy}{dx} = xy \sin x$. **(6 marks)**

**11** **a** Find $\int \frac{3x + 4}{x}\,dx$, $x > 0$. **(2 marks)**

**b** Given that $y = 16$ at $x = 1$, solve the differential equation $\frac{dy}{dx} = \frac{3x\sqrt{y} + 4\sqrt{y}}{x}$ giving your answer in the form $y = f(x)$. **(6 marks)**

# Homework Exercise

**12** **a** Express $\dfrac{8x-18}{(3x-8)(x-2)}$ in partial fractions. **(3 marks)**

**b** Given that $x \geqslant 3$, find the general solution to the differential equation

$$(x-2)(3x-8)\frac{dy}{dx} = (8x-18)y$$ **(5 marks)**

**c** Hence find the particular solution to this differential equation that satisfies $y = 8$ at $x = 3$, giving your answer in the form $y = f(x)$. **(4 marks)**

**13** **a** Find the general solution of $\dfrac{dy}{dx} = 2x - 4$.

**b** On the same axes, sketch three different particular solutions to this differential equation.

**14** **a** Find the general solution to the differential equation $\dfrac{dy}{dx} = -\dfrac{1}{(x+2)^2}$ **(3 marks)**

**b** On the same axes, sketch three different particular solutions to this differential equation. **(3 marks)**

**c** Write down the particular solution that passes through the point (8, 3.1). **(1 mark)**

**15** **a** Show that the general solution to the differential equation $\dfrac{dy}{dx} = -\dfrac{x}{y}$ can be written in the form $x^2 + y^2 = c$. **(3 marks)**

**b** On the same axes, sketch three different particular solutions to this differential equation. **(3 marks)**

**c** Write down the particular solution that passes through the point (0, 7). **(1 mark)**

# Homework Answers

**1** **a** $y = A\,e^{x-x^2} - 1$ **b** $y = k \sec x$

**c** $y = \dfrac{-1}{\tan x - x + c}$ **d** $y = \ln|2e^x + c|$

**2** **a** $\dfrac{1}{24} - \dfrac{\cos^3 x}{3}$ **b** $\sin 2y + 2y = 4\tan x - 4$

**c** $\tan y = \frac{1}{2}\sin 2x + x + 1$ **d** $y = \arccos(e^{-\tan x})$

**3** **a** $y = Axe^{-\frac{1}{x}}$ **b** $y = -\,e^3 xe^{-\frac{1}{x}} = -\,xe^{\left(\frac{3x-1}{x}\right)}$

**4** $y = \sqrt{\dfrac{x}{x+1}}$ **5** $\ln|2 + e^y| = -x\,e^{-x} - e^{-x} + c$

**6** $y = \dfrac{3}{1-x}$ **7** $y = \dfrac{3(1+x^2)+1}{3(1+x^2)-1}$

**8** $y = \ln\left|\dfrac{x^2-12}{2}\right|$ **9** $\tan y = x + \dfrac{1}{2}\sin 2x + \dfrac{2-\pi}{4}$

**10** $\ln|y| = -\,x\cos x + \sin x - 1$

**11** **a** $3x + 4\ln|x| + c$ **b** $y = \left(\dfrac{3}{2}x + 2\ln|x| + \dfrac{5}{2}\right)^2$

**12** **a** $\dfrac{5}{3x-8} + \dfrac{1}{x-2}$

**b** $\ln|y| = \frac{5}{3}\ln|3x-8| + \ln|x-2| + c$

**c** $y = 8(x-2)(3x-8)^{\frac{5}{3}}$

**13** **a** $y = x^2 - 4x + c$

**b** Graphs of the form $y = x^2 - 4x + c$, where $c$ is any real number

**14** **a** $y = \dfrac{1}{x+2} + c$

**b** Graphs of the form $y = \dfrac{1}{x+2} + c$, where $c$ is any real number

**c** $y = \dfrac{1}{x+2} + 3$

**15** **a** $\dfrac{dy}{dx} = -\dfrac{x}{y} \Rightarrow \int y\,dy = \int -x\,dx$

$\Rightarrow \frac{1}{2}y^2 = -\frac{1}{2}x^2 + b \Rightarrow y^2 + x^2 = c$

**b** Circles with centre (0, 0) and radius $\sqrt{c}$, where $c$ is any positive real number.

**c** $y^2 + x^2 = 49$